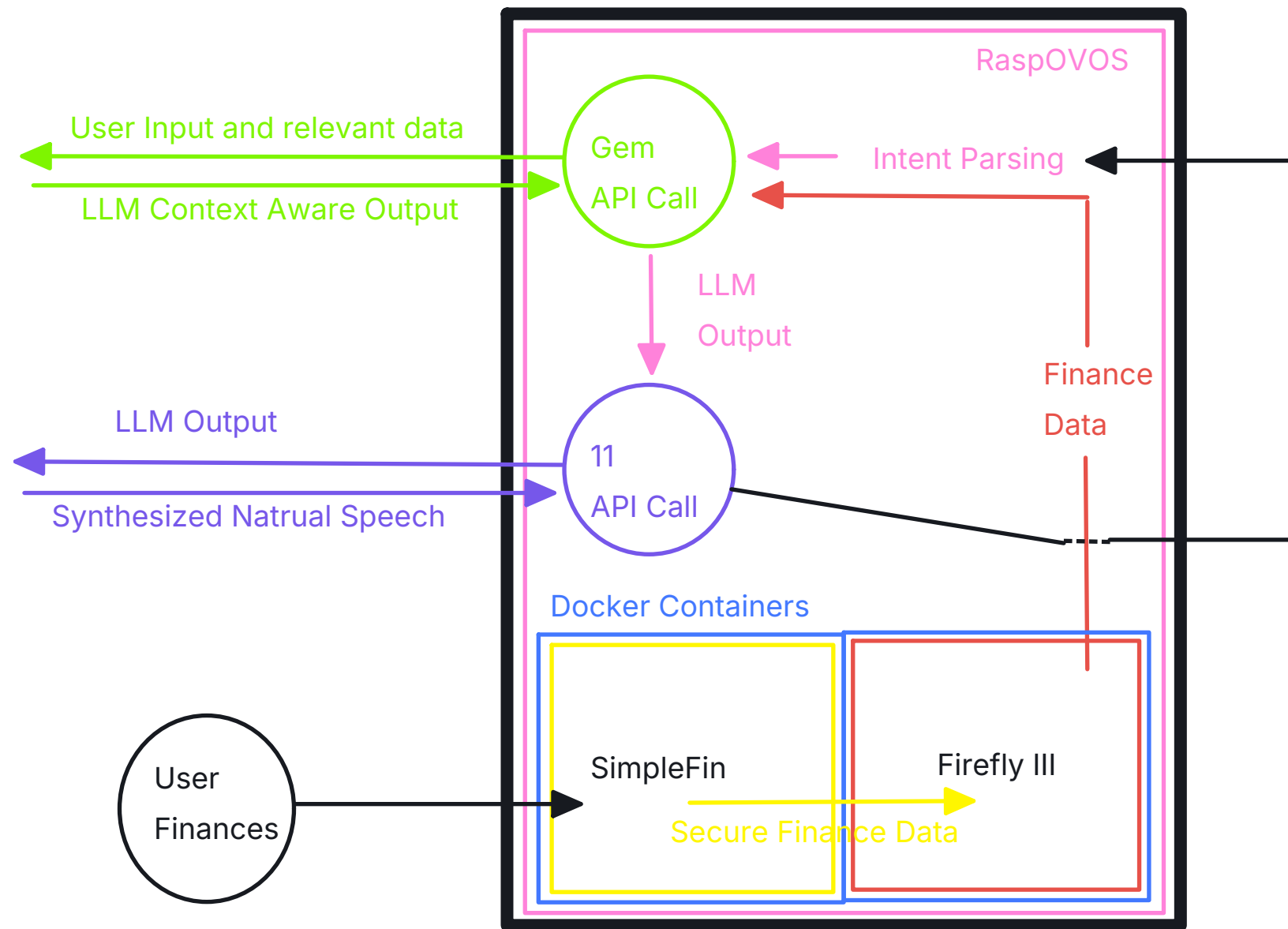


A voice assistant designed to provide the user with information regarding their financial accounts and provide context aware budgeting advice based on the user's current finances.

Software Stack

- **RaspOVOS** - Raspberry Pi focused fork of OVOS for processing user speech and performing speech to text conversation. Enables responses to call relevant APIs when an input containing certain context relevant keywords is received.
- **Docker** - Run sensitive financial data aggregation services in isolated docker images to ensure security.
- **Firefly III** - Personal finance management software, this contains the user's financial data in this case and acts as the personal account the program is pulling data from while we originally planned to use the Nessie API for this task we've been told the API is down currently so we're using this in its place.
- **SimpleFIN** - Secure financial data aggregator, allows the transfer of financial data without exposing sensitive information or providing direct access to financial accounts.
- **Google Gemini API** - LLM to process user requests and provide context aware responses based on the restraints and data defined in the API setup. This also allows the use of adjacent google services to retrieve needed data for questions about budget sensitive shopping, financial institutions, etc.
- **ElevenLabs** - Text to speech API used to make device responses to user input sound natural and likable. We want the device to act as a companion which means we need responses that feel natural and personable.

System Topology



Major Tasks

1. Set up docker container with Firefly III and SimpleFIN, ensure compatibility with RaspOVOS installation and if no issues are found or issues are able to be corrected in an acceptable amount of time continue with dev
2. Create read-only token from SimpleFIN for OVOS to query financial data with.
3. Develop OVOS skills for the software to be able to query financial data and make API calls to Gemini and ElevenLabs
4. Refine Gemini API call skill to ensure the LLM is provided the proper system

prompting, this step will be repeated to fine tune the LLM responsiveness unless we somehow get it first try then hooray i guess.

5. Develop the function to call the ElevenLabs API with the Gemini output and await the returned natural speech text to speech response, play this back for the user.

Tasks to be explored if major tasks are in an acceptable state

1. Explore advanced functionality like comparing lending options or comparing financial institutions (Note: This is assuming that the data this service uses is procured independently of a financial institution i.e. being treated as a third party service rather than one provided by a bank.
2. Voice customizability vis LLM response
3. Include a screen over I2C for a fun face to make the device more personable (needs to be picked up from UH campus first)
4. Explore local options for voice synthesis to reduce cloud reliance (check if ElevenLabs has any edge models)
5. Try to integrate conversational interactions.